INCH-POUND

MS51842E
10 March 2017
SUPERSEDING
MS51842D
30 July 1987

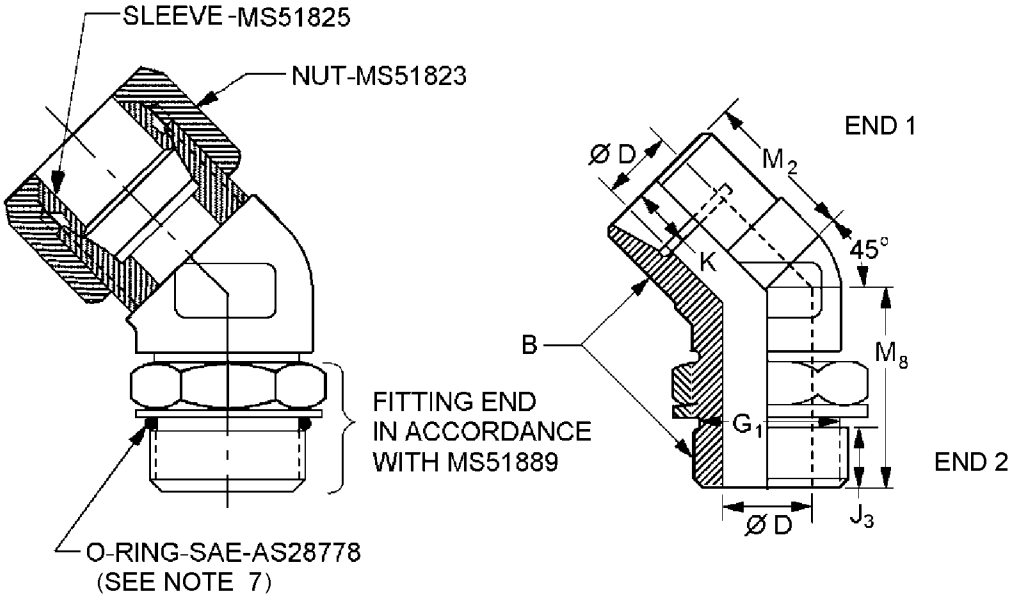
DETAIL SPECIFICATION SHEET

ELBOW, TUBE TO BOSS 45 DEGREE, O-RING, FLARELESS TYPE, HYDRAULIC

Reactivated after 10 March 2017 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-18866.



See notes at end of figure.

FIGURE 1. Elbow, tube to boss 45°.

AMSC N/A

FSC 4730

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Dash number (see note 5)		Tube OD Nom inches (mm)	B Straight thread Class 2A	D Diameter Drill inches (mm)	G_1 Diameter inches (mm) +.002 (0.05) -.003 (0.08)
Assembly	Body				
-1	-21	.1250 (3.175)	.3125 - 24UNF	.093 (2.36)	.250 (6.35)
-2	-22	.1875 (4.763)	.3750 - 24UNF	.125 (3.18)	.313 (7.95)
-3	-23	.2500 (6.350)	.4375 - 20UNF	.203 (5.16)	.364 (9.25)
-4	-24	.3125 (7.936)	.5000 - 20UNF	.234 (5.94)	.427 (10.85)
-5	-25	.3750 (9.525)	.5625 - 18UNF	.281 (7.14)	.482 (12.24)
-6	-26	.5000 (12.700)	.7500 - 16UNF	.422 (10.72)	.660 (16.76)
-7	-27	.6250 (15.875)	.8750 - 14UNF	.500 (12.70)	.773 (19.63)
-8	-28	.7500 (19.050)	1.0625 - 12UN	.656 (16.66)	.945 (24.00)
-9	-29	.8750 (22.225)	1.1875 - 12UN	.718 (18.24)	1.070 (27.18)
-10	-30	1.0000 (25.400)	1.3125 - 12UN	.875 (22.23)	1.195 (30.35)
-11	-31	1.2500 (31.750)	1.6250 - 12UN	1.093 (27.76)	1.507 (38.28)
-12	-32	1.5000 (38.100)	1.8750 - 12UN	1.344 (34.14)	1.756 (44.60)
-13	-33	2.0000 (50.800)	2.5000 - 12UN	1.813 (46.05)	2.381 (60.48)

Dash number (see note 5)		J_3 Full thread ±.005 (0.13)	K inches (mm) +.015 (0.38) -.005 (0.13)	M_2 inches (mm) ±.03 (0.8)	M_8 inches (mm) ±.03 (0.8)
Assembly	Body				
-1	-21	.234 (5.94)	.188 (4.78)	.64 (16.26)	.86 (21.84)
-2	-22	.234 (5.94)	.234 (5.94)	.64 (16.26)	.86 (21.84)
-3	-23	.281 (7.14)	.234 (5.94)	.70 (17.78)	1.05 (26.67)
-4	-24	.281 (7.14)	.250 (6.35)	.73 (18.54)	1.05 (26.67)
-5	-25	.312 (7.92)	.250 (6.35)	.83 (21.08)	1.14 (28.96)
-6	-26	.344 (8.74)	.305 (7.75)	.98 (24.89)	1.30 (33.02)
-7	-27	.391 (9.93)	.350 (8.89)	1.08 (27.43)	1.52 (38.61)
-8	-28	.469 (11.91)	.350 (8.89)	1.27 (32.26)	1.73 (43.94)
-9	-29	.469 (11.91)	.350 (8.89)	1.27 (32.26)	1.80 (45.72)
-10	-30	.469 (11.91)	.415 (10.54)	1.36 (34.54)	1.86 (47.24)
-11	-31	.469 (11.91)	.415 (10.54)	1.45 (36.83)	1.91 (48.51)
-12	-32	.469 (11.91)	.485 (12.32)	1.52 (38.61)	1.91 (48.51)
-13	-33	.469 (11.91)	.485 (12.32)	1.83 (46.48)	1.91 (48.51)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Remove all burrs and slivers.
4. Dimensions and tolerances not shown shall be in accordance with SAE-J514.
5. Assemblies are only furnished to this specification (body, nut, and sleeve). Bodies are not to be stocked stored or issued.
6. The drawing is for identification purposes only and is not intended to restrict designs and shapes not dimensioned.
7. Procure the O-ring separately and install immediately before installation of the fitting.

FIGURE 1. Elbow, tube to boss 45° - Continued.

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REQUIREMENTS:

Elbow, tube to boss 45° shall be as specified on figure 1 and in table I.

Materials shall be in accordance with MIL-DTL-18866.

Finish. Finishes shall be as specified in table I. All plating's shall be capable of meeting a minimum of 96 hours salt spray test in accordance with ASTM B117. The fittings shall show no evidence of red corrosion after 96 hours of salt spray. Fluid passages, other openings and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.

TABLE I. Material and finish identification codes.

PIN code material/plating finish	Material	Plating finish
Blank	Steel	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2. 1/
H	Steel	Zinc-Aluminum in accordance with ASTM F1136/F1136M, grade 3, NC. 2/
J	Steel	Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 2/ 3/
M	Nickel-copper alloy UNS N04400	No additional finish.
N	High-chromium nickel alloy UNS N06690	No additional finish.
P	Steel	Zinc phosphate finish in accordance MIL-DTL-16232 type Z, class 1. 4/
R	Steel	Zinc plating in accordance with ASTM B633; type III or V, Fe/Zn 8. 4/
S	Corrosion resistant steel	No additional finish. Passivation in accordance with SAE-AMS2700, method 1, type 6 or 7.
T	Titanium	Anodize in accordance with SAE-AMS2488 type 2. 5/
TF	Titanium	Fluoride phosphate in accordance with SAE-AMS2486. 5/
Z	Steel	Zinc plating in accordance with ASTM B695, type II, class 8 4/
ZC	Steel	Zinc may be any zinc plating from PIN codes J, R, and Z with a colored chromate coating. 4/

1/ Embrittlement test need not be run. Cadmium shall not be used in oxygen or potable water systems.
2/ Hexavalent chromium free. Finish shall be ROHS compliant.
3/ The zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315µ inches (8µm) minimum.
4/ Not for use in aircraft. Requires approval from the Program Office for all applications.
5/ A pretreatment, a modification of the fluoride treatment, or a post treatment shall be applied so the final color of the fittings shall be similar to SAE-AMS-STD-595 colors 36076 through 36293. Titanium shall not be used in oxygen or potable water systems.

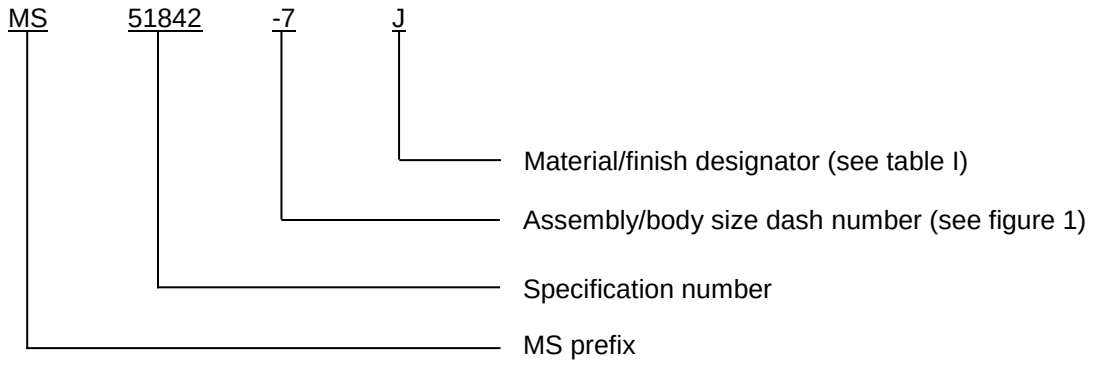
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Trivalent wrenchability. When the finish has been damaged due to poor wrenchability, the surface of the connector shall be touched up using one of the brush plating processes below as appropriate to primary finish. The term “trivalent wrenchability” is used to evaluate the ability of the finish to withstand abrasion from an excessive amount of wrenching.

a. Brush plating of hard chromium by electrodeposition shall be in accordance with SAE-AMS-2451/5.

b. Brush plating of medium-hardness, low stress nickel by electrodeposition shall be in accordance with SAE-AMS-2451/9.

Maximum operating pressure. Maximum operating pressure shall be in accordance with SAE-J514.
PIN: The PIN consists of the letters “MS”, the specification number, a dash, number for an elbow, tube to boss 45°, assembly/body size, and a letter for material finish designator.



PIN example: MS51842-7J indicates an elbow, tube to boss 45°, end1 is .6250 inch (15.875 mm) tube OD, end 2 is a .8750 - 14UNF-2A thread, steel with zinc-nickel plating.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

Class I and II ozone depleting substances (ODS) shall not be used in MS51842 or any referenced procedures.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

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Referenced documents. In addition to MIL-DTL-18866, this document references the following:

MIL-DTL-16232	SAE-AMS-STD-595/36081	SAE-AMS-STD-595/36280
MS51823	SAE-AMS-STD-595/36099	SAE-AMS-STD-595/36293
MS51825	SAE-AMS-STD-595/36118	SAE-AMS2417
MS51889	SAE-AMS-STD-595/36134	SAE-AMS2451/5
ASTM B117	SAE-AMS-STD-595/36152	SAE-AMS2451/9
ASTM B633	SAE-AMS-STD-595/36170	SAE-AMS2486
ASTM B695	SAE-AMS-STD-595/36173	SAE-AMS2488
ASTM F1136/F1136M	SAE-AMS-STD-595/36176	SAE-AMS2700
SAE-AMS-C-81562	SAE-AMS-STD-595/36231	SAE-AS28778
SAE-AMS-QQ-P-416	SAE-AMS-STD-595/36251	SAE-J514
SAE-AMS-STD-595/36076	SAE-AMS-STD-595/36270	

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - SH
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2017-046)

Review activities:
Army - AR
Navy - CG, MC, SA
Air Force - 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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